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1  /*****
2  * Longitudinal Data Analysis – Assignment 3
3  * Alzheimer CDRSB binary outcome: GEE + GLMM
4  *****/
5  /*****
6  * 0. Set library
7  *****/
8  libname eddy "/home/u64020665";
9
10 /*****
11 * 1. Construct long-format dataset + create binary outcome YBIN
12 *****/
13 data eddy.alz_long2;
14     set eddy.alz_wide;
15
16     array bprs_arr[7] bprs0-bprs6;
17     array cdrsb_arr[7] cdrsb0-cdrsb6;
18
19     do time = 0 to 6;
20
21         time2 = time*time;    /* quadratic time term */
22
23         BPRS = bprs_arr[time+1];
24         CDRSB = cdrsb_arr[time+1];
25
26         /* Missingness indicators (Q1) */
27         miss_BPRS = (BPRS = .);
28         miss_CDRSB = (CDRSB = .);
29
30         /* -----
31          Binary outcome (Q3): define YBIN from CDRSB
32          CHANGE THIS to match your Assignment 2 definition.
33          Example below: YBIN = 1 if CDRSB >= 4, else 0
34          ----- */
35         if CDRSB = . then YBIN = .;
36         else YBIN = (CDRSB >= 4);
37
38         output;
39     end;
40
41     keep patid trial sex age edu bmi inkomen job adl wzc
42         abpet0 taupet0
43         time time2 BPRS CDRSB YBIN miss_BPRS miss_CDRSB;
44 run;
45
46 /*****
47 Sort and create the response indicator
48 *****/
49 proc sort data=eddy.alz_long2 out=alz_s;
50     by patid time;
51 run;
52
53 data ipw_dat;
54     set alz_s;
55     R = (YBIN ne .);    /* 1=observed, 0=missing */
56 run;
57
58 data ipw_dat;
59     set ipw_dat;
60     by patid;
61
62     YBIN_lag = lag(YBIN);
63     if first.patid then YBIN_lag = .;

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64  /* If you want, also add lag of BPRS or CDRSB */
65  /* Example:
66  CDRSB_lag = lag(CDRSB);
67  if first.patid then CDRSB_lag = .;
68  */
69  run;
70
71  /*****
72  Fit the observed model
73  *****/
74
75  proc logistic data=ipw_dat;
76    class sex / param=ref;
77    model R(event='1') =
78      time age sex bmi adl abpet0 taupe0
79      YBIN_lag; /* include history (recommended) */
80    output out=ipw_pi p=pi;
81  run;
82
83  /*****
84  Construct inverse probability weights
85  *****/
86
87  proc logistic data=ipw_dat;
88    model R(event='1') = time;
89    output out=ipw_num p=pi_num;
90  run;
91
92  proc sort data=ipw_pi; by patid time; run;
93  proc sort data=ipw_num; by patid time; run;
94
95  data ipw_w;
96    merge ipw_pi ipw_num;
97    by patid time;
98
99    /* Avoid division by 0 */
100    pi = max(pi, 1e-6);
101    pi_num = max(pi_num, 1e-6);
102
103    w = pi_num / pi; /* stabilized weight */
104
105    /* Keep only rows where YBIN is observed (GEE uses observed outcomes) */
106    if R=1;
107  run;
108
109  /*****
110  Fit the weighted GEE
111  *****/
112
113  data ipw_w;
114    set ipw_w;
115    time2 = time*time;
116  run;
117
118  proc genmod data=ipw_w descending;
119    class patid sex;
120    model YBIN =
121      time time2
122      age sex bmi adl
123      abpet0 taupe0
124      / dist=binomial link=logit;
125    weight w;
126    repeated subject=patid / type=exch corrw;
127  run;
128
129  proc means data=ipw_w n mean std min p1 p5 p50 p95 p99 max;

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128   var w;
129 run;
130
131 /*****
132 Reestimating weights
133 *****/
134
135 proc sort data=eddy.alz_long2 out=alz_s;
136   by patid time;
137 run;
138
139 data ipw_dat;
140   set alz_s;
141   R = (YBIN ne .);
142   time2 = time*time;
143 run;
144
145 /* Denominator: P(R=1 | baseline covariates + time) */
146 proc logistic data=ipw_dat;
147   class sex / param=ref;
148   model R(event='1') = time time2 age sex bmi adl abpet0 taupet0;
149   output out=ipw_pi p=pi;
150 run;
151
152 /* Numerator: P(R=1 | time) (stabilization) */
153 proc logistic data=ipw_dat;
154   model R(event='1') = time;
155   output out=ipw_num p=pi_num;
156 run;
157
158 proc sort data=ipw_pi; by patid time; run;
159 proc sort data=ipw_num; by patid time; run;
160
161 data ipw_w2;
162   merge ipw_pi ipw_num;
163   by patid time;
164
165   pi      = max(pi,      1e-4); /* slightly larger floor than 1e-6 to avoid explosions */
166   pi_num = max(pi_num, 1e-4);
167
168   w = pi_num / pi;
169
170   if R=1; /* keep observed outcomes only */
171 run;
172
173 /*****
174 Trim weights
175 *****/
176
177 proc univariate data=ipw_w2 noprint;
178   var w;
179   output out=wcap pctlpts=99 pctlpre=P_;
180 run;
181
182 data _null_;
183   set wcap;
184   call symputx('wcap99', P_99);
185 run;
186
187 data ipw_w2_trim;
188   set ipw_w2;
189   w_trim = min(w, &wcap99);
190 run;
191
192 proc means data=ipw_w2_trim n mean std min p50 p95 p99 max;
193   var w w_trim;

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192 run;
193
194 proc genmod data=ipw_w2_trim descending;
195     class patid sex;
196     model YBIN =
197         time time2
198         age sex bmi adl
199         abpet0 taupet0
200         / dist=binomial link=logit;
201     weight w_trim;
202     repeated subject=patid / type=exch corrw;
203 run;
204
205 /*****
206 *****/
207 /*****
208 Multiple imputation
209 *****/
210 /*****
211 FCS with logistic regression
212 *****/
213
214 proc mi data=eddy.alz_long2
215     out=mi_dat
216     seed=20251230
217     nimpute=10;
218     class patid sex YBIN;
219     fcs logistic(YBIN = time time2 age sex bmi adl abpet0 taupet0);
220     var patid time time2 YBIN age sex bmi adl abpet0 taupet0;
221 run;
222
223
224 /*****
225 Fit GEE
226 *****/
227
228 ods output GEEEmpPEst = gee_mi_pe;
229
230 proc genmod data=mi_dat descending;
231     by _Imputation_;
232     class patid sex;
233     model YBIN =
234         time time2
235         age sex bmi adl
236         abpet0 taupet0
237         / dist=binomial link=logit;
238     repeated subject=patid / type=exch;
239 run;
240
241 ods output close;
242
243 /*****
244 Prepare results
245 *****/
246
247 data gee_mi_ready;
248     set gee_mi_pe;
249
250     length Parameter $40;
251     Parameter = lowercase(Parm);
252     if Parm = "sex" then Parameter = cats("sex=", Level1);
253
254     keep _Imputation_ Parameter Estimate StdErr;
255 run;

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256 /*****
257 Pool estimates
258 *****/
259 .....
260 proc sort data=gee_mi_ready;
261   by Parameter;
262 run;
263 .....
264 proc mianalyze data=gee_mi_ready;
265   by Parameter;
266   modeleffects Estimate;
267   stderr StdErr;
268 run;
269 .....
270 ods output ParameterEstimates = mi_pooled;
271 .....
272 proc mianalyze data=gee_mi_ready;
273   by Parameter;
274   modeleffects Estimate;
275   stderr StdErr;
276 run;
277 .....
278 ods output close;
279 .....
280 proc print data=mi_pooled;
281 run;
282 /*****
283 Prediction grid
284 *****/
285 .....
286 /* Create prediction grid */
287 proc means data=eddy.alz_long2 noprint;
288   var age bmi adl abpet0 taupe0;
289   output out=means mean=age bmi adl abpet0 taupe0;
290 run;
291 .....
292 data pred_grid;
293   set means;
294   do time = 0 to 6;
295     time2 = time*time;
296     sex = 0; /* reference category */
297     output;
298   end;
299   keep time time2 age sex bmi adl abpet0 taupe0;
300 run;
301 .....
302 proc genmod data=ipw_final descending;
303   class patid sex;
304   model YBIN = time time2 age sex bmi adl abpet0 taupe0
305     / dist=binomial link=logit;
306   weight w_trim;
307   repeated subject=patid / type=exch;
308   store out=ipwgee_store;
309 run;
310 .....
311 /*-----
312 0) Build IPW_FINAL (stable + trimmed weights)
313 -----*/
314 .....
315 proc sort data=eddy.alz_long2 out=alz_s;
316   by patid time;
317 run;
318 .....
319 data ipw_dat;
320   set alz_s;

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320 R = (YBIN ne .);
321 run;
322
323 /* Denominator model */
324 proc logistic data=ipw_dat;
325     class sex / param=ref;
326     model R(event='1') = time time2 age sex bmi adl abpet0 taupet0;
327     output out=ipw_pi p=pi;
328 run;
329
330 /* Numerator model (stabilization) */
331 proc logistic data=ipw_dat;
332     model R(event='1') = time;
333     output out=ipw_num p=pi_num;
334 run;
335
336 proc sort data=ipw_pi; by patid time; run;
337 proc sort data=ipw_num; by patid time; run;
338
339 data ipw_w2;
340     merge ipw_pi ipw_num;
341     by patid time;
342
343     pi = max(pi, 1e-4);
344     pi_num = max(pi_num, 1e-4);
345
346     w = pi_num / pi;
347
348     if R=1;
349 run;
350
351 /* Trim at 99th percentile */
352 proc univariate data=ipw_w2 noprint;
353     var w;
354     output out=wcaps pctlpts=99 pctlpre=P_;
355 run;
356
357 data _null_;
358     set wcaps;
359     call symputx('wcap99', P_99);
360 run;
361
362 data ipw_final;
363     set ipw_w2;
364     w_trim = min(w, &wcap99);
365 run;
366
367 /* Verify ipw_final exists and contains the needed variables */
368 proc contents data=ipw_final; run;
369
370 /*-----
371 1) Fit IPW-GEE and STORE
372 -----*/
373
374 proc genmod data=ipw_final descending;
375     class patid sex;
376     model YBIN = time time2 age sex bmi adl abpet0 taupet0
377         / dist=binomial link=logit;
378     weight w_trim;
379     repeated subject=patid / type=exch;
380     store out=ipwgee_store;
381 run;
382
383 proc plm restore=ipwgee_store;
384     score data=pred_grid out=pred_ipw predicted=prob_ipw / ilink;
385 run;
386

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384 proc plm restore=ipwgee_store;
385     score data=pred_grid
386         out=pred_ipw
387         predicted=prob_ipw
388         / ilink;
389 run;
390
391 proc genmod data=mi_dat descending;
392     by _Imputation_;
393     class patid sex;
394     model YBIN =
395         time time2 age sex bmi adl abpet0 taupet0
396         / dist=binomial link=logit;
397     repeated subject=patid / type=exch;
398     store out=mi_gee_store;
399 run;
400
401 proc plm restore=mi_gee_store;
402     score data=pred_grid
403         out=pred_mi_all
404         predicted=prob_mi
405         / ilink;
406 run;
407
408 proc means data=pred_mi_all noprint;
409     class time;
410     var prob_mi;
411     output out=pred_mi mean=prob_mi;
412 run;
413
414 proc sort data=pred_ipw; by time; run;
415 proc sort data=pred_mi; by time; run;
416
417 data pred_compare;
418     merge pred_ipw(keep=time prob_ipw)
419           pred_mi (keep=time prob_mi);
420 by time;
421 run;
422
423 proc print data=pred_compare;
424 run;
425
426 proc sgplot data=pred_compare;
427     series x=time y=prob_ipw /
428         lineattrs=(thickness=2)
429         legendlabel="IPW-GEE";
430
431     series x=time y=prob_mi /
432         lineattrs=(thickness=2 pattern=dash)
433         legendlabel="MI-GEE";
434
435     yaxis label="Predicted probability of YBIN = 1";
436     xaxis label="Time" integer;
437     keylegend / location=inside position=topright;
438 run;
439
440
441 /* Refit IPW-GEE and store model */
442 proc genmod data=ipw_w2_trim descending;
443     class patid sex;
444     model YBIN =
445         time time2 age sex bmi adl abpet0 taupet0
446         / dist=binomial link=logit;
447     weight w_trim;

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448 repeated subject=patid / type=exch;
449 store out=ipwgee_store;
450 run;
451
452 /* Score IPW-GEE predictions */
453 proc plm restore=ipwgee_store;
454 score data=pred_grid out=pred_ipw predicted=prob_ipw / ilink;
455 run;
456
457 proc genmod data=mi_dat descending;
458 by _Imputation_;
459 class patid sex;
460 model YBIN =
461     time time2 age sex bmi adl abpet0 tauPET0
462     / dist=binomial link=logit;
463 repeated subject=patid / type=exch;
464 store out=mi_gee_store;
465 run;
466
467 proc plm restore=mi_gee_store;
468 score data=pred_grid out=pred_mi_all predicted=prob_mi / ilink;
469 run;
470
471 proc means data=pred_mi_all noprint;
472 class time;
473 var prob_mi;
474 output out=pred_mi mean=prob_mi;
475 run;
476
477 proc sort data=pred_ipw; by time; run;
478 proc sort data=pred_mi; by time; run;
479
480 data pred_compare;
481 merge pred_ipw(keep=time prob_ipw)
482       pred_mi (keep=time prob_mi);
483 by time;
484 run;
485
486 proc sgplot data=pred_compare;
487 series x=time y=prob_ipw / lineattrs=(thickness=2) legendlabel="IPW-GEE";
488 series x=time y=prob_mi / lineattrs=(thickness=2 pattern=dash)
489     legendlabel="MI-GEE";
490 yaxis label="Predicted probability of YBIN=1";
491 xaxis label="Time";
492 keylegend / location=inside position=topright;
493 run;
494
495
496
497
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499
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